FARHAN MOHAMMAD KHAN, Ph. D.

Research Associate

Department of Civil Engineering, Birla Institute of Technology and Science, Pilani (BITS Pilani), India

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EDUCATION

Qualification	Institute	Specialization	Percentage	Year
Ph.D.	Birla Institute of Technology and Science, Pilani (RJ) *Thesis title:* Physico-Chemical Parameters Based Artificial Intelligence Model for Prediction of Faecal Coliform Bacteria in Ground Water *Thesis Supervisor:* Prof. Rajiv Gupta, Senior Professor, Dept. of Civil Engineering, BITS Pilani.	Environmental Engineering	-	2019- 2022
M.E	Samrat Ashok Technological Institute (Autonomous), Vidisha (M.P) *Thesis title:* Grey Water & Kitchen Waste Management for SATI (Campus), Vidisha *Thesis Supervisor:* Prof. A. K. Saxena, Dept. of Civil Engineering, SATI Vidisha	Environmental Engineering	7.94 CGPA	2016-2018
PGDCA	AISECT University, Bhopal (M.P)	Computer Applications	77.1%	2016-2017
B.E	Mahakal Institute of Technology & Management, Ujjain (M.P)	Civil Engineering	7.3 CGPA	2011-2015
XII	Shri Guru Teg Bahadur Public Higher Secondary School, Ratlam (M.P)	PCM	68%	2011
X	Shri Guru Teg Bahadur Public Higher Secondary School, Ratlam (M.P)	All compulsory subjects	79.8%	2009
DOA	Softech Computer Centre, Ratlam (M.P.)	Computer Applications	PAAS	2007

TECHNICAL SKILLS

- Computational & Programming: Python, MATLAB, AutoCAD, ArcGIS, Minitab, and MS Office.
- Artificial Intelligence & Machine learning, Internet of Things (IoT) & Smart Sensors, Geospatial Analysis & GIS Applications, Academic & Scientific Writing, Journal Selection & Peer Review Expertise, Research Proposal Writing & Grant Applications.
- Water & Wastewater Analysis: pH, ORP, EC, TSS, TDS, TOC, Turbidity, DO, BOD, COD, Alkalinity, Hardness,Chlorides, Sulphates, Sodium, Potassium, Nitrates, Total Phosphates, Fluoride, Heavy Metals, Total Coliform, Feacal Coliform, Oil & Grease, etc.
- Artificial Intelligence & Machine learning, Internet of Things (IoT) & Smart Sensors, Geospatial Analysis & GIS

Applications, Academic & Scientific Writing, Journal Selection & Peer Review Expertise, Research Proposal Writing & Grant Applications

- Soil Analysis: CEC, ESP, Sodium, Potassium, Calcium, Magnesium, Bulk Density, Porosity, Water Holding Capacity, Organic Carbon, Heavy Metals, etc.
- Instrument Handling & Data Analysis: Atomic Absorption Spectrophotometer, Inductively Coupled Plasma - Optical Emission Spectrophotometer, UV- Spectrophotometer, Flame Photometer, Optical Microscope, KR Box, pH Meter, Conductivity Meter, Turbidity Meter, Autoclave, BOD Incubator, Labtronics LT-51 Water & Soil Testing Kit, YSI ProDSS Handheld Multiparameter Water Quality Meter, RaspBerry Pi, and Arduino UNO.
- Operating Systems: Microsoft Windows, Ubuntu, and macOS.

AREA OF INTEREST

Artificial Intelligence, Environmental, Social and Governance, Geographic Information System, Environmental Microbiology, Water Quality, Environmental Impact Assessment, Groundwater Contamination, Wastewater Engineering, Water Treatment, Urban Water Management.

WORK EXPERIENCE

Birla Institute of Technology and Science, Pilani *February 2023 - Till now*

Research Associate, Department of Civil Engineering

Project 1: Structured Dialogues for Sustainable Urban Water Management (SDSUWM) sponsored by the Department of Science and Technology (DST) under the file number DST/TM/EWO/WTI/2K19/UWS-04.

- To assess the urban impact and dynamics (social, equity, uncertainties, risk, resilience, environment, economic) on water system.
- To assess, map, and analyse the unequal access to water-related infrastructure as socio-spatial, socio-culture, regulation-governance, historically followed water management schemes at ground level
- Integration of opinions and solution of stakeholders and consortium for Infrastructure projects planning, design, construction, and management using Cradle to Cradle LCA for sustainable implementation

Project 2: Design and Development of a Smart Device for Detecting Water Quality in Municipal Water Distribution Systems.

- Developing a low-cost system for real time monitoring of water quality.

Paperpedia Pvt Ltd, Gurgaon *October 2022 - January 2022*

Academic Content Writer

CSIR-National Environmental Engineering Research Institute, Nagpur *December 2021 - September 2022*

Senior Project Associate, Environmental Impact and Sustainability Division

Project 1: Carrying Capacity Study for Critically Polluted Areas (CPAs), Severely Polluted Areas (SPAs) and Other Polluted Area (OPAs) listed under Comprehensive Environmental Pollution Index in Maharashtra.

Locations: Aurangabad, Nashik, Chandrapur, Nagpur, Dombivli, Navi Mumbai

Roles and Responsibilities:

- Ground Water Sampling, Assessment, and Report Preparation.
- Surface Water Sampling, Assessment, and Report Preparation.
- Soil Sampling, Assessment, and Report Preparation.
- Air Quality Monitoring.
- Noise Quality Monitoring.
- Socio-economic Survey.

Project 2: Preparation of Environmental Impact Assessment Report for Kalpasar Dam
Roles and Responsibilities:

- Ground Water Sampling, Assessment, and Report Preparation
- Sediments and Benthos Sampling
- Marine Water Sampling, Assessment, and Report Preparation
- Air Quality Monitoring
- Noise Quality Monitoring
- Socio-economic Survey

Additional Projects and Responsibilities:

- Preparation of Standard Operating Procedures (SOPs) for Air Quality Monitoring
- Preparation of Environmental Impact Assessment Study Report for the Proposed Petrochemical Complex along with Marine Jetty at Cuddalore, Tamilnadu as per EIA 2006 Notification.
- Preparation of Environmental Sustainability Report for Grasilene Division – Harihar Grasim Industries Ltd. (Pulp & Fibre Business).

Birla Institute of Technology and Science, Pilani *January 2019 – December 2021*
Junior Research Fellow, Department of Civil Engineering
Project: Costs and Remediation of Groundwater Contamination in India.

- Water Sampling, Testing and Report Preparation.
- Automating the process of bacterial detection using Artificial Intelligence.
- Automating the process of bacterial colony counting using Artificial Intelligence.
- Preparation and Distribution of Water Filters for Fluoride Removal in Ground Water.
- Prediction of fluoride in nails using Artificial Intelligence to identify the degree of fluoride exposure.

Subjects Taught, Course: UG

- Environmental Engineering.
- Applications of AI in Civil Engineering.
- Remote Sensing and GIS.
- Fluid Mechanics.

Royal College of Technology, Ratlam *September 2018 – December 2018*
Assistant Professor, Department of Civil Engineering
Subjects Taught, Course: Diploma

- Environmental Engineering.
- Surveying
- Construction Materials
- Quantity Surveying and Costing
- Basic Civil Engineering and Engineering Mechanics

Shri Yogindra Sagar Institute of Technology and Science, Ratlam *July 2015– July 2016*
Assistant Professor, Department of Civil Engineering
Subjects Taught, Course: UG

- Building Planning and Architecture
- Engineering Geology.
- Water Resources Engineering.
- Environmental Engineering.

- Environmental Chemistry.
- Engineering Hydrology.

ACADEMIC PROJECTS

- M.E Thesis Project: Grey Water & Kitchen Waste Management for SATI (Campus), Vidisha.
- B.E Technical Projects: Minor project on Water Treatment Plant, Major project on Vermicomposting, Seminar on Flexible Pavement, Railway Sleeper, Demolition of building, Moisture Resisting Roads.

ACHIEVEMENTS & AWARDS

- Attended and delivered an oral presentation at the *International Conference on Sustainable Urban Water Management for Fast-Growing Cities*, held from 23rd to 25th October 2024 at BITS Pilani, Hyderabad. Received *First Prize* for *Best Oral Presentation Award*.
- Received the Best Research Award in 2020 from ScienceFather for the article titled "ARIMA and NAR based prediction model for time series analysis of COVID-19 cases in India," published in the Journal of Safety Scienceand Resilience, 1(1), 12-18 (Khan & Gupta, 2020).
- Attended and participated in the poster competition at the Symposium on Carbon Nanomaterial Electronics (SCNE 2019) held at BITS Pilani on November 8th-9th, 2019. It has been selected as one of the best poster competitions.
- Attended and delivered an oral presentation at the 2019 6th International Conference on Environmental Research(ICESR 2019) in Melbourne, Australia, from 18th to 20th December 2019. It has been selected as one of the bestoral presentations.
- Participated and won First prize in Writing Competition at the school level.
- Participated and won First prize in Running Race at the district level.
- Participated and won Bronze Medal in Football at the college level.

PUBLICATIONS

JOURNALS

- Kumar, A., Khan, F. M., & Gupta, R. (2024). Spatial Analysis of Water Quality Trends in Wastewater Treatment Using GIS and Machine Learning. In *World Environmental and Water Resources Congress 2024* (pp. 1451-1470).
- Nidheesh, P. V., Khan, F. M., Kadier, A., Akansha, J., Bote, M. E., & Mousazadeh, M. (2022). Removal of nutrients and other emerging inorganic contaminants from water and wastewater by electrocoagulation process. *Chemosphere*, 135756.
- Khan, F. M., Gupta, R., & Sekhri, S. (2021). Automated Bacteria Colony Counting on Agar Plates Using MachineLearning. *Journal of Environmental Engineering (ASCE)*, *147*(12), 04021066. https://doi.org/10.1061/(ASCE)EE.1943-7870.0001948**.**
- Khan, F. M., Gupta, R., & Sekhri, S. (2021). A novel PCA-FA-ANN based hybrid model for prediction offluoride. *Stochastic Environmental Research and Risk Assessment (Springer)*, 1-28. https://doi.org/10.1007/s00477-021-02001-4
- Khan, F. M., Gupta, R., & Sekhri, S. (2021). Superposition learning-based model for prediction of E. coli in groundwater using physico-chemical water quality parameters. *Groundwater for Sustainable Development (Elsevier)*, Volume 13, 2021, 100580. https://doi.org/10.1016/j.gsd.2021.100580.
- Khan, F. M., Gupta, R., & Sekhri, S. (2021). A convolutional neural network approach for detection of E. coli bacteria in water. *Environmental Science and Pollution Research (Springer)*, 1-9. https://doi.org/10.1007/s11356-021-14983-3
- Khan, F. M., Gupta, R., & Sekhri, S. (2021). Analysis of increase in cell counts of Escherichia coli in groundwaterof Rajasthan: Possible presence of VBNC cells. *Research Journal of Chemistry and Environment*, Vol, 25, 6.
- Khan, F. M., Saxena, A. K. (2017). Integrated Approach towards Grey Water Management, Treatment, and Reuse. *IJSRD*, 5(7), 691-693.
- Kumar, G., Kumar, A., Khan, F. M., & Gupta, R, Sprawl of the COVID-19 in changing scenario: a methodologybased on

social interaction, 2021, Library Hi Tech. https://doi.org/10.1108/LHT-06-2020-0134**.** https://doi.org/10.1016/j.jnlssr.2020.06.007.

- Khan, F. M., Kumar, A., Puppala, H., Kumar, G., & Gupta, R. (2021). Projecting the Criticality of COVID-19 Transmission in India Using GIS and Machine Learning Methods. *Journal of Safety Science and Resilience*. https://doi.org/10.1016/j.jnlssr.2021.05.001.
- Lounis, M., & Khan, F. M. (2021). Predicting COVID-19 cases, deaths and recoveries using machine learning methods. *Engineering and Applied Science Letter, 4(4)*, 43-49.
- Khan, F. M., & Gupta, R. (2020). ARIMA and NAR based prediction model for time series analysis of COVID- 19 cases in India. *Journal of Safety Science and Resilience, 1(1)*, 12-18. ***Best Research Award***
- Kumar, A., Khan, F. M., Gupta, R., & Puppala, H. (2020). Preparedness and mitigation by projecting the riskagainst COVID-19 transmission using machine learning techniques. *medRxiv*. https://doi.org/10.1101/2020.04.26.20080655.
- Khan, F., Saxena, A.K., & Kushwaha, A. (2018). Sewage/Wastewater Treatment Technologies with Special Reference to Oxidation Pond: A Tool For Waste Water Treatment. *International journal of scientific research in science, engineering and technology, 4*, 817-830.

BOOK

- Khan, F. M. (2023). Artificial Intelligence Model for Prediction of Faecal Coliform Bacteria in Ground Water.ISBN: 9785406065259.

BOOK CHAPTER

- Khan, F. M., & Gupta, R. (2020). Escherichia coli (*FCB*) as an Indicator of Fecal Contamination in Groundwater:A Review. In *International Conference on Sustainable Development of Water and Environment* (pp. 225-235). Springer, Cham, The 3rd International Conference on Sustainable Development of Water and Environment (ICSDWE 2020), 13-14 January, 2020, Inha University, South Korea. https://doi.org/10.1007/978-3-030-45263- 6_21.

CONFERENCE

- Khan, F. M., Sridhar, S., & Gupta, R. (2020). Detection of waterborne bacteria using Adaptive Neuro-Fuzzy Inference System. In *E3S Web of Conferences* (Vol. 158, p. 05002). EDP Sciences, 6th International Conference on Environmental Systems Research (ICESR 2019), 18-20 December, 2019, Melbourne, Australia. https://doi.org/10.1051/e3sconf/202015805002. ***Best paper award***

PATENT

- Portable Hand-Held Smart device for real-time Water Quality Measurement and Water Quality Classification, Abheek Gupta, Farhan Mohammad Khan, Anu Gupta, Rajiv Gupta, Application No.: 202111017453; Filed on 14.4.2021; India.

TRAINING CONDUCTED

- Guest Speaker, Expert Talk on "Sustainable Environmental Management Along with Artificial Intelligence" at the Department of Civil Engineering (IET), SAGE University, Indore, delivering a short-term training program (STTP) on 19/05/2023.

ACTIVE EDITOR OF THE FOLLOWING JOURNALS/CONFERENCE PROCEEDINGS

- Acta Scientific.
- Plos ONE.
- Futuristic Trends in Construction Materials & Civil Engineering, Volume 3, Book 1, 2024, IIP Series.

ACTIVE REVIEWER OF THE FOLLOWING JOURNALS/CONFERENCE PROCEEDINGS

- Egyptian Journal of Basic and Applied Sciences (Taylor and Francis).

- Egyptian Journal of Aquatic Research (Elsevier).
- International Conference on Applied Mathematics and Epidemics Analysis, DZMath’21, 21-22 Sept 2021.
- The Journal of Life and Environmental Sciences (PeerJ).
- Environment, Development and Sustainability (Springer).
- Environmental Science and Pollution Research (Springer).
- Acta Scientific.
- Cogent Environmental Science (Taylor and Francis).
- Human Vaccines & Immunotherapeutics.
- Journal of Intelligent Systems.
- Information Discovery and Delivery.
- International Journal of Environmental Science and Development.

PERSONAL DETAILS

Date of Birth: 04/04/1994.
Address: 156, Ved Vyas Colony No.4, Ratlam (M.P). Pin Code: 457001.
Father’s Name: Farhat Mohd. Khan (M.A, LLB, B.Ed.), Occupation: Retired Government Officer.
Mother’s Name: Akhtar Khan (B.Sc., M.Sc., B.Ed.).
Brother’s Name: Adnan Mohd. Khan (B. Tech Computer Science), Occupation: Software Engineer, Infosys Limited, Pune.

DECLARATION

I hereby declare that the above information is correct to the best of my knowledge and belief.

FARHAN M. KHAN